US Semiconductors

Quantum State of the Union: focus on fault tolerance fundamentals, not funding fears

Industry Overview

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Equity
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Remain constructive on quantum; focus on roadmaps

IONQ saw pressure after all-stock $1.3bn Elisity and $2bn stock offering heightened dilution concern. IonQ's fundraising has raised questions around why the company is raising add'l cash ahead of next-gen Superion platform's 2Q27 launch with ~$2bn of cash already on the balance sheet post SKYT. While no immediate use of capital has been noted, we see pot'l for IonQ to expand quantum technology portfolio via continued industry investment with minimal further fundraising expected in n/m-t. DARPA moved IONQ into QBI Stage C, worth up to another $300m and an important endorsement of tech capability. QNT's selloff may have been driven by its absence from first wave of Stage C vendors, though we note each vendor has different timelines independent of quantum platform performance/commercial viability. QNT continues to have a significant logical qubit lead vs. peers, while next-gen Sol (2H27) and Apollo (2029) platforms remain on track. Quantum stocks are expected to remain high-beta, though trapped-ion remains our preferred modality as we maintain Buy on IONQ and QNT and remain constructive on their ability to generate quantum advantage before 2030. We also attended OCP's quantum track, which featured Diraq, Nu Quantum and IQM.

IONQ: Stage C plus $2bn funds Superion roadmap to FT

IonQ is one of four new Stage C entrants (Atom, Diraq, IBM). Evaluators will test several Superion gens through '29, with funding beyond initial tranche subject to appropriations. IonQ's equity offering was priced at $40, came with 100m 7-yr warrants, and net proceeds were ~$2bn. We see the raise as providing additional dry powder for Skywater investment, roadmap R&D, and pot'l future M&A while derisking roadmap progression.

IONQ: Elisity adds security as a second growth engine

IonQ is acquiring Elisity for up to $1.3bn in stock, expected to close in 4Q. We est. Elisity generates ~$62m ARR based on recent security multiples. Elisity's posture mgmt. maps customer networks, identifying pot'l vulnerabilities - important given rising AI/quantum cyber concerns. IonQ would be able to monetize the mapping in addition to QKD hardware followed by cybersecurity subscriptions thereafter. The adoption of post quantum cryptography (PQC) methods is not expected to drive significant revenue (standards are open-source, much like RSA-2048). However, as the capabilities of quantum increase, we expect the importance of QKD in addition to PQC to increase, creating a l-t growth engine for IonQ that other quantum peers have little exposure to.

QNT: Stage C is about timing; fault-tolerance on hardware

QNT remains in Stage B. DARPA says QBI does not pick "winners and losers" and expects more Stage C promotions "in the near future." QNT remains the quantum performance leader, while Helix (QNT's error-correction architecture) running on Helios beat the physical baseline on every benchmark without discarding any runs and demonstrated industry leading logical performance. Recent JPM-QNT work demonstrated near-break-even performance of complex, error corrected algorithmic quantum circuits using only fault tolerant components. Sol (2H27) and Apollo (2029) remain on track while QNT has $2.1bn of cash and limited dilution risk given its compute-only focus.

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Refer to important disclosures on page 4 to 8. Analyst Certification on page 3. Price Objective Basis/Risk on page 2. **13029005**

Exhibit 1: Stocks mentioned
Prices and ratings for stocks mentioned in this report

BofA Ticker	Bloomberg ticker	Company name	Price	Rating
IONQ	IONQ US	IonQ Inc	US$ 34.35	C-1-9
QNT	QNT US	Quantinuum	US$ 36.68	C-1-9

Source: BofA Global Research

BofA GLOBAL RESEARCH

Glossary

ARR: Annual Recurring Revenue
DARPA: Defense Advanced Research Projects Agency
FT: Fault-Tolerant
IONQ: IonQ
JPM: JPMorgan Chase
OCP: Open Compute Project
PQC: Post-Quantum Cryptography
QBI: Quantum Benchmarking Initiative
QKD: Quantum Key Distribution
QNT: Quantinuum
RSA: Rivest–Shamir–Adleman (public-key encryption)
SKYT: SkyWater Technology

Price objective basis & risk

IonQ Inc (IONQ)
We have a $60 SOTP-based PO on IonQ. Relative to quantum peers, IonQ's revenue scale, customer engagements and logical-qubit scaling potential are partially offset by roadmap execution and M&A integration risk. As such, we apply 7.0x FY30E EV/sales to $2.5bn of Quantum Hardware & Services revenue, in line with the peer median, and 2.0x to $1.2bn of Foundry revenue (slightly ahead of peers given higher growth), plus $2.0bn of estimated post-SkyWater cash. This implies a 5.4x blended FY30E EV/sales multiple. We use FY30E sales because IonQ remains loss-making with negative FCF through FY30E.

Upside risks are: 1) earlier Superion delivery and logical-qubit milestones, 2) stronger customer conversion and system acceptance, 3) faster manufacturing iteration or better yield, and 4) stronger-than-expected Foundry growth.

Downside risks are: 1) Superion delays or failure to demonstrate fully error-corrected logical qubits on an integrated commercial system, 2) slower quantum adoption and customer conversion, 3) competitive progress across quantum modalities, 4) M&A, SkyWater integration, utilization, yield and capital-intensity risk, 5) customer concentration, uneven revenue recognition and lower Foundry margins, and 6) higher cash burn, dilution and long-duration valuation sensitivity.

Quantinuum (QNT)
Our $85 price objective is based on 8x our 2030E EV/Sales. We use 2030E EV/Sales given QNT revenue remains subscale with profit not expected until FY30E and valuation is tied to roadmap execution, logical-qubit scaling and customer conversion. Our multiple is above most relevant trapped-ion peer IonQ trading at 5x but justified given QNT's superior logical qubit count, fidelity of current commercial system, and derisked roadmap.

Upside risks are faster Sol/Apollo/Lumos execution, earlier fault-tolerant quantum adoption, stronger cloud/on-prem utilization, larger government/sovereign awards and better software/services monetization.

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Downside risks are roadmap delays, limited customer ROI, weaker TAM realization, pilot projects failing to convert to recurring revenue, intensifying competition across trapped-ion, superconducting, neutral-atom and photonic modalities, potential export-control/government ownership constraints, and high stock volatility given reliance on 2030+ revenue estimates.

Analyst Certification

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US - Semiconductors and Semiconductor Capital Equipment Coverage Cluster

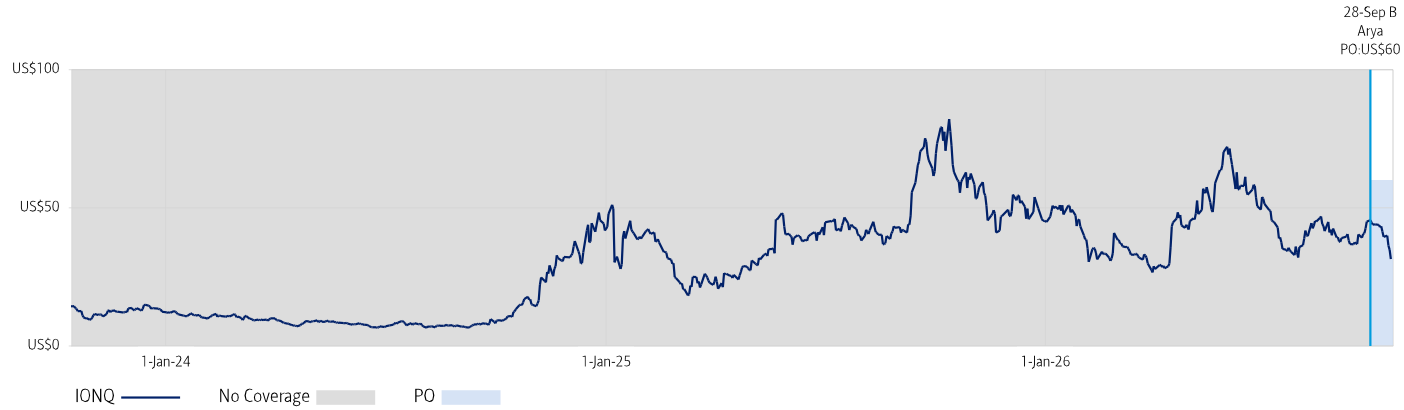
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BUY				
	Advanced Energy Industries	AEIS	AEIS US	Duksan Jang
	Advanced Micro Devices, Inc	AMD	AMD US	Vivek Arya
	Allegro Microsystems	ALGM	ALGM US	Vivek Arya
	Analog Devices Inc.	ADI	ADI US	Vivek Arya
	Applied Materials, Inc.	AMAT	AMAT US	Vivek Arya
	Broadcom Inc	AVGO	AVGO US	Vivek Arya
	Cadence	CDNS	CDNS US	Vivek Arya
	Camtek	CAMT	CAMT US	Michael Mani
	Credo Technology	CRDO	CRDO US	Vivek Arya
	Diodes Inc.	DIOD	DIOD US	Michael Mani
	Intel	INTC	INTC US	Vivek Arya
	IonQ Inc	IONQ	IONQ US	Vivek Arya
	KLA Corporation	KLAC	KLAC US	Vivek Arya
	Lam Research Corp.	LRCX	LRCX US	Vivek Arya
	M/A-Com	MTSI	MTSI US	Vivek Arya
	Marvell Technology, Inc.	MRVL	MRVL US	Vivek Arya
	Microchip	MCHP	MCHP US	Vivek Arya
	Micron Technology, Inc	MU	MU US	Vivek Arya
	MKS Instruments	MKSI	MKSI US	Michael Mani
	Nova	NVMI	NVMI US	Michael Mani
	NVIDIA Corporation	NVDA	NVDA US	Vivek Arya
	onsemi	ON	ON US	Vivek Arya
	Quantinuum	QNT	QNT US	Vivek Arya
	Synopsys	SNPS	SNPS US	Vivek Arya
	Teradyne	TER	TER US	Vivek Arya
	Texas Instruments Inc.	TXN	TXN US	Vivek Arya
NEUTRAL				
	Ambarella	AMBA	AMBA US	Vivek Arya
	Ambiq Micro, Inc.	AMBQ	AMBQ US	Vivek Arya
	Arm Holdings	ARM	ARM US	Vivek Arya
	Astera Labs Inc	ALAB	ALAB US	Vivek Arya
	Coherent Corp	COHR	COHR US	Vivek Arya
	Lumentum Holdings	LITE	LITE US	Vivek Arya
	NXP Semiconductors NV	NXPI	NXPI US	Vivek Arya
UNDERPERFORM				
	Axcelis Technologies	ACLS	ACLS US	Duksan Jang
	GlobalFoundries	GFS	GFS US	Vivek Arya
	Lattice Semiconductor	LSCC	LSCC US	Duksan Jang
	Qualcomm	QCOM	QCOM US	Vivek Arya
	Skyworks Solutions, Inc.	SWKS	SWKS US	Vivek Arya
RVW				
	Wolfspeed Inc	WOLF	WOLF US	Vivek Arya

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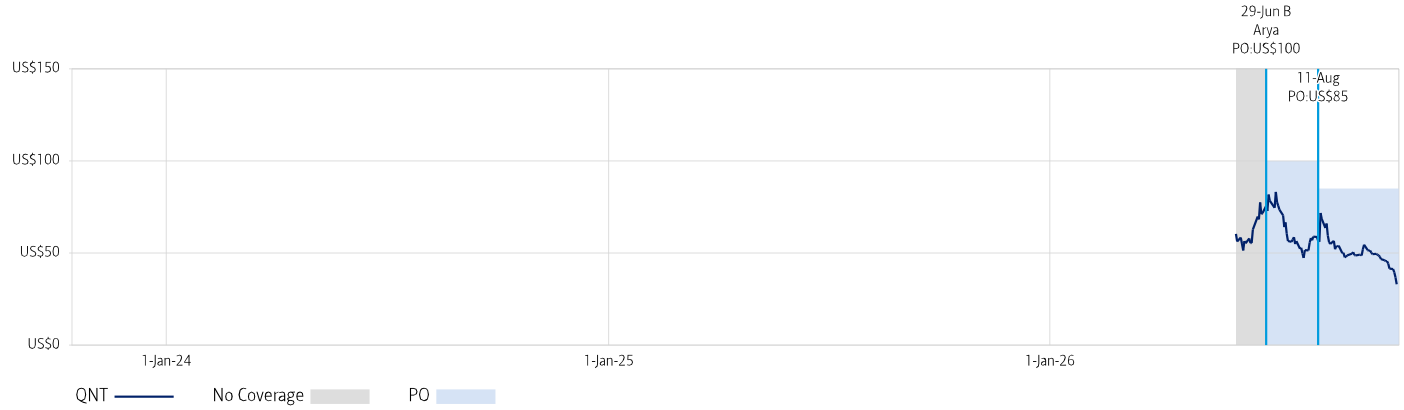
IonQ Inc (IONQ) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Quantinuum (QNT) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

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Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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